BIO 005 HUMAN PHYSIOLOGY · INSTRUCTOR ONLY

Clicker key: Week 10, Pressure, Flow, and Blood Pressure

Three class sets, picked on the deck's title bar. Concept clickers: 30 s, confidence points. Label-it clickers: 15 s, one tap. Rooms VESSELS10, VESSELS10B, VESSELS10C on the dashboard.

SET A · ROOM VESSELS10 · KEY: DBCABADCDCAB			ANSWER	
1	CONCEPT	An arteriole's radius falls to half. With the same pressure gradient, what happens to flow through it?	D	Falls to one sixteenth
2	LABEL	Name the structure at the pin.	B	Vein
3	CONCEPT	A cuff reading is 130/70. What is the pulse pressure and the mean arterial pressure?	C	60 and 90
4	LABEL	Name the structure at the pin.	A	Mean arterial pressure
5	CONCEPT	During a run, the arterioles in the leg muscles dilate even though sympathetic activity is high. What explains the dilation?	B	Local metabolites (CO_2, H^+, K^+, adenosine, low O_2) override sympathetic tone
6	CONCEPT	At the venous end of a capillary, hydrostatic pressure is 15 mm Hg and plasma colloid osmotic pressure is 25 mm Hg. Which way does fluid move, and how strongly?	A	In, 10 mm Hg
7	LABEL	Name the structure at the pin.	D	Capillary wall (endothelium with pores)
8	CONCEPT	A guard stands motionless for an hour and faints. Trace the cause.	C	Blood pooled in the leg veins, venous return and stroke volume fell, MAP fell, brain flow fell
9	CONCEPT	Blood pressure falls when a patient stands. In the next few seconds, which combination does the baroreflex produce?	D	Less vagal firing, more sympathetic firing: HR up, contractility up, arterioles and veins constrict
10	LABEL	Name the structure at the pin.	C	Vagus nerve (CN X), afferent
11	LABEL	Name the structure at the pin.	A	Aldosterone (adrenal cortex)
12	CONCEPT	During moderate exercise, total peripheral resistance falls yet mean arterial pressure rises slightly. How?	B	Cardiac output rises more than resistance falls

SET B · ROOM VESSELS10B · KEY: ACDBCABDBCAD			ANSWER	
1	CONCEPT	Which vessel is built to be a reservoir, and what feature makes it one?	A	Vein; thin compliant wall and large lumen
2	LABEL	Name the structure at the pin.	C	Pulse pressure in the artery
3	CONCEPT	While deflating a cuff you hear the first tap at 118 and the sounds disappear at 76. What was happening in the artery at 118?	D	The pulse peak first exceeded cuff pressure, so blood spurted through in a turbulent burst
4	LABEL	Name the structure at the pin.	B	Last Korotkoff sound
5	CONCEPT	Arterial pressure rises suddenly, yet flow to the kidney barely changes. Which mechanism did that?	C	The myogenic response: stretch made the arterioles constrict
6	CONCEPT	A patient with liver failure has swollen ankles and ascites. Which Starling force is abnormal?	A	Plasma colloid osmotic pressure is down because albumin is low
7	CONCEPT	Why do deep breaths raise venous return?	B	Thoracic pressure falls and abdominal pressure rises, pulling and pushing blood toward the chest
8	LABEL	Name the structure at the pin.	D	Respiratory pump: thoracic pressure falls on inspiration

SET B · ROOM VESSELS10B · KEY: ACDBCABDBCAD			ANSWER	
9	CONCEPT	An ACE inhibitor is started. Which step stops, and what two downstream effects fall?	B	Angiotensin I to angiotensin II; arteriolar constriction and aldosterone fall
10	LABEL	Name the structure at the pin.	C	Sympathetic output to the heart
11	LABEL	Name the structure at the pin.	A	Atrial natriuretic peptide (ANP)
12	CONCEPT	Two hours after losing a liter of blood, a patient's hematocrit has fallen even though no fluid has been given. Why?	D	Interstitial fluid moved into the capillaries because capillary pressure fell and absorption exceeded filtration

SET C · ROOM VESSELS10C · KEY: DACBACDBCBDA			ANSWER	
1	CONCEPT	Blood velocity is slowest in the capillaries even though they are the narrowest vessels. Why?	D	Their total cross-sectional area is the largest, and velocity = flow ÷ area
2	LABEL	Name the structure at the pin.	A	Precapillary sphincter
3	CONCEPT	A patient's diastolic pressure rises from 80 to 100 with little change in systolic. Which variable most likely changed?	C	Total peripheral resistance
4	LABEL	Name the structure at the pin.	B	Diastolic pressure
5	CONCEPT	Nitroglycerin relieves chest pain by giving the vessel nitric oxide. Which pathway does it switch on?	A	NO → cGMP → Ca^{2+} down and phosphatase on → relax
6	CONCEPT	After lymph nodes are removed from one armpit, that arm slowly swells. Why?	C	The arm's lymphatics can no longer return the 3 L a day of leftover filtrate and leaked protein
7	LABEL	Name the structure at the pin.	D	Net filtration (arterial end)
8	CONCEPT	A vein's valves have failed (varicose veins). Which pump stops working properly?	B	Skeletal muscle pump, because blood squeezed upward falls back down
9	CONCEPT	A patient's atria are stretched by fluid overload. Which hormone rises, and what does it do?	C	Atrial natriuretic peptide; the kidney excretes Na^+ and water and renin is inhibited
10	LABEL	Name the structure at the pin.	B	Carotid sinus baroreceptor
11	LABEL	Name the structure at the pin.	D	Angiotensin I
12	CONCEPT	Why does shock become irreversible past a certain point even if the lost blood is replaced?	A	Underperfused heart and brain centers fail, acidosis relaxes arterioles and capillaries leak, so the fall in pressure feeds itself